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United States Patent	12414537
Kind Code	B1
Date of Patent	September 16, 2025
Inventor(s)	Teper; Brett et al.

Litter box with removable liner

Abstract

A litter box includes a container having a wall and pegs that project upward from a top of the wall or horizontally from an inner surface of the wall, the plurality of pegs being distributed along a circumference of the container. A liner is insertable into the container with tabs connected along an upper edge of the liner. Each tab includes a connection part and a stiffener part. The connection part extends upward from the upper edge of the liner and has an opening in which a respective peg of the pegs is received to hold the liner in position in the container. The stiffener part of each tab extends laterally from opposing sides of the upper portion and holds the liner against the wall of the container when the tab is connected to the respective peg.

Inventors:	Teper; Brett (Brooklyn, NY), Williams; Richard (Brooklyn, NY)
Applicant:	ModProducts, LLC (Brooklyn, NY)
Family ID:	1000008039592
Assignee:	MODPRODUCTS, LLC (Brooklyn, NY)
Appl. No.:	18/780708
Filed:	July 23, 2024

Related U.S. Application Data

continuation-in-part parent-doc US 29932824 20240315 PENDING child-doc US 18780708

Publication Classification

Int. Cl.:	A01K1/01 (20060101)
U.S. Cl.:	
CPC	A01K1/0107 (20130101);

Field of Classification Search

CPC: A01K (1/0107); A01K (1/0157); B65F (1/065)

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Primary Examiner: Huson; Joshua D

Assistant Examiner: Peterson; Alanna K

Attorney, Agent or Firm: LUCAS & MERCANTI, LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a Continuation-in-Part of U.S. patent application Ser. No. 29/932,824, filed on Mar. 15, 2024, the entire contents of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

(1) The present invention relates to a litter box with a removable liner, the liner including tabs for suspending and holding the liner against sides of the litter box.

(2) Prior art liners are used in litter boxes to keep the litter box clean and also can be used as

garbage bags to remove the soiled litter. In some cases the prior art liners can be emptied and reused. In either case, it is important to prevent litter, urine, or any type of debris from getting between the liner and the walls of the litter box. One way to accomplish this is to drape over the top rim of the litter box. However, this is not an aesthetically pleasing solution.

(3) In one known solution, the liner includes eyelets or grommets that attached to hooks which are hung from the rim of the litter box. However, the hooks can slide along the rim and create gaps between the liner and the walls of the litter box.

BRIEF SUMMARY OF THE INVENTION

(4) An object of the present invention is to provide a liner for a litter box that is easily removable and can be held in place along a circumference of the top of the liner to prevent litter, urine, or any other type of debris from getting between the liner and the litter box.

(5) The object is met by a litter box including a container having a wall and pegs that project upward from a top of the wall or horizontally from an inner surface of the wall, the plurality of pegs being distributed along a circumference of the container. The litter box further includes a liner that is insertable into the container with tabs connected along an upper edge of the liner. Each tab includes an upper portion and a lower portion. The upper portion extends upward from the upper edge of the liner and has an opening in which a respective peg of the pegs is received to hold the liner in position in the container. The lower portion of each tab extends laterally from opposing sides of the upper portion and holds the liner against the wall of the container when the tab is connected to the respective peg.

(6) In one embodiment, the container has four wall portions forming four corners, the four wall portions including a front wall portion, a rear wall portion, a first side wall portion, and a second side wall portion, and the plurality of pegs comprises four pegs arranged so that the plurality of tabs connected to the four pegs holds the liner in the four corners of the container. Although this embodiment includes four pegs and four tabs, other embodiments may use more or less than four pegs and four tabs to hold the liner against the inner wall of the container.

(7) In a preferred embodiment, a first pair of the four pegs is arranged on the first side wall portion and a second pair of the four pegs is arranged on the second side wall portion. The four pegs are disposed so that one and of the lower portion of each tab of the plurality of tabs holds the liner in each of the four corners of the container.

(8) In another embodiment of the invention, the four corners are curved sections, and the four pegs are disposed at the curved sections of the four corners.

(9) In a further embodiment, the plurality of pegs project upward from the top of the wall, and the each tab of the plurality of tabs includes a living hinge on the upper portion, by which the opening of the each tab can be pivoted to fit over a corresponding peg of the plurality of pegs. The upper portion of the each tab is divided into a first part and a second part by the living hinge, the first part being disposed between the living hinge and the lower portion and the second part including the opening. While this embodiment describes the pegs as projecting upward, the pegs may also project in a direction having a horizontal component as well as a vertical component, i.e., upward and toward an outside of the container or upward and toward an inside of the container.

(10) Each tab may include a rib disposed on at least a portion of a border of the opening to reinforce the opening and prevent it from damage.

(11) Each tab may also include a grip portion extending from a free end of the each tab to make it easier for a user to place the tab on a peg or remove the tab from the peg.

(12) According to another embodiment of the invention, the plurality of pegs project horizontally from the inner surface of the wall of the container. The each peg of the plurality of pegs includes a groove proximate a base thereof in which a portion of a corresponding tab of the plurality of tabs is received to prevent the corresponding tab from slipping off of the each peg. Alternatively, or additionally, each peg of the plurality of pegs includes a rib proximate a base thereof behind which a portion of a corresponding tab of the plurality of tabs is received to prevent the corresponding tab

from slipping off of the each peg. Although this embodiment describes the pegs as projected horizontally, the pegs may also project in a direction having a vertical component, i.e., toward an inside of the container and upward.

(13) The lower portion of the each tab is connected to the liner by at least one of stitching, friction fitting, gluing, welding, pinning, riveting, snap-fitting, and crimping.

(14) In a preferred embodiment, the lower portion of each tab is sewn into a corresponding pocket of the liner.

(15) In yet another embodiment, the tab is incorporated into the liner as a die-cut portion of the material of the liner.

(16) Other objects and features of the present invention will become apparent from the following detailed description considered in conjunction with the accompanying drawings. It is to be understood, however, that the drawings are designed solely for purposes of illustration and not as a definition of the limits of the invention, for which reference should be made to the appended claims. It should be further understood that the drawings are not necessarily drawn to scale and that, unless otherwise indicated, they are merely intended to conceptually illustrate the structures and procedures described herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In the drawings, wherein like reference characters denotes similar elements throughout the several views:

(2) FIG. 1 is a perspective view of a litter box according to an embodiment of the present invention;

(3) FIG. 2 is a perspective view of a liner of the litter box in FIG. 1;

(4) FIG. 3 is a closer view of area A in FIG. 1;

(5) FIG. 4 is a perspective view of a tab of the liner in FIG. 2;

(6) FIG. 5 is a side view of the tab in FIG. 4;

(7) FIG. 6 is a perspective view of a litter box according to another embodiment of the present invention;

(8) FIG. 7 is a perspective view of a liner of the litter box in FIG. 6;

(9) FIG. 8 is a perspective view of a section of the litter box in FIG. 6 shown without the liner;

(10) FIG. 9 is a side view of the horizontal peg of FIG. 8;

(11) FIG. 10 is a perspective view of the section of the litter box in FIG. 8 with the liner inserted; and

(12) FIG. 11 is a perspective view of a tab according to the embodiment of FIGS. 6-10.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(13) FIG. 1 shows an embodiment of a litter box **100** according to a first embodiment of the present invention. The litter box **100** includes a container **110** and a liner **116**. The container **110** has an opening **118** on one of the wall sections and the liner **116** has a corresponding opening. A cover **114** is placed on top of the container **110**. An upper rim **111** of the container **110** includes pegs **112** that project upward.

(14) FIG. 2 shows the liner **116** used in the container **110** of FIG. 1. The liner **116** has a shape that corresponds to the shape of the container **110** and is inserted into the container **110** as shown in FIG. 1. The liner **116** includes tabs **120** that extend upward from the upper edge of the liner **116**. The tabs **120** correspond to respective ones of the pegs **112** and define holes through which the pegs **112** can be received when the liner **116** is placed in the container **110**. The connection between the tabs **120** and the pegs **112** suspends and holds the liner **116** in place in the container **110**.

(15) As shown in FIGS. 3-5, the tab **120** includes a stiffener part **122** that braces a portion of the

liner against an inner wall of the container and a connection part **124** that connects the liner **116** to a peg **112** on the container **110**. The stiffener part **122** is sewn into a top portion of the liner **116**. In the specific embodiment shown in FIGS. **1** and **2**, a top of the liner is folded over and the stiffener part **122** is disposed between the folded parts with the connection part **124** passed through a hole in the liner **116**. The folded parts of the liner **116** are then sewn to form a pocket in the top portion of the liner **116** that holds the stiffener part **122**. Although the stiffener part **122** is shown as being sewn in a pocket of the liner **116**, any other fixing techniques such as, but not limited to, stitching, friction fitting, gluing, welding, pinning, riveting, snap-fitting, and clinching may be used to fix the stiffener part **122** to the liner **116**.

(16) The connection part **124** includes a first portion **126** connected to the stiffener part **122** and a second portion **127** connected to the first portion **126** by a living hinge **125**. When the liner **116** is placed in the container **110**, the second portion **127** is pivoted approximately 90° relative to the first part **126** about the living hinge **125**. The living hinge **125** allows the liner **116** to be packaged and shipped in a flat form. The second portion **127** is pivoted relative to the first portion **126** when the liner **116** is ready to be placed in the container **110**. The second portion **127** is then placed over one of the pegs **112**. In an alternative embodiment, the living hinge may be omitted and the second portion **127** always extends 90° relative to the first part **126**.

(17) The first portion **126** allows a top of the liner **116** to be held at a distance from the top of the container **110**, i.e., the distance between the hinge **125** and the stiffener part **122**. However, the first portion **126** may be omitted so that the second portion **127** of the connection part **124** is connected directly to the lower tab part **122** with or without the living hinge **125**.

(18) The stiffener part **122** acts as a brace and holds the top of the liner **116** against the inner wall of the container **110**. The stiffener part **122** extends laterally on opposing sides of the connection part **124**. In a preferred embodiment, one side of the stiffener part **122** extends to a position proximate a corner of the container **110** to hold the top of the liner **116** taut against the container **110**.

(19) The second portion **127** of the connection part **124** includes a rib **128** on both a front side and a back side that extends along the border of the hole through which the peg **112** is received. The rib **128** reinforces the hole.

(20) The second portion **127** also includes a grip portion **129** that extends from the free end of the connection part **124**. The grip portion **129** allows a user to grasp the second portion **127** for pivoting second portion **127** relative to the first portion **126** about the living hinge **125**. The grip also facilitates removing the liner **116** from the container **110** when required.

(21) The entire tab **120** may be made as a monolithic element made of one uniform material, i.e., polypropylene, or another plastic. However, the tab is not limited to a plastic and may be made of other materials such as cardboard or a metal. As a further alternative, the connection part **124** and stiffener part **122** may be made of two different materials. For example, the stiffener part **122** may be made of metal and the connection part **124** made of plastic. In this case, the stiffener part **122** and the connection part **124** could be connected by a molding process.

(22) FIG. **6** shows another embodiment of a litter box **200** with a container **210** that includes an opening **218** on one of the wall sections. The liner **216** has a corresponding opening. The container **210** includes pegs **212** that extend horizontally from an inner wall of the container **210** and the liner **216** includes tabs **220** configured to be received on the pegs to suspend and hold the liner **216** in place in the container **210**. An additional clip **225** may be used at the bottom of the opening **218** to secure the liner **216** to the container **210** to the opening **218**.

(23) FIG. **7** shows the liner **216** that is used in the litter box **200**. In this embodiment, the pegs **212** are arranged in rounded corners of the container **210** and the stiffeners **256** (shown in FIG. **11**) of the tabs **220** bend to match the curve of the container **210**. Although four pegs **212** and four tabs **220** are shown, any number of pegs and tabs may be used depending on the size and shape of the container **210** and the material of the liner **216**, or any other factors that affect the ability of the liner to be held against the inner wall of the container.

(24) The horizontal pegs **212** are shown in more detail in FIGS. **8** and **9**. In this embodiment, the tabs **220** are not required to have a living hinge. To help maintain the tabs **220** on the horizontal peg **212**, a groove **244** or channel may be formed at the base end of the peg **212** in which a portion of the tab **220** is received. The tab **220** must be lifted out of the groove **244** to remove the tab **220** from the peg **212**. Further, a rib **242** is placed in front of the groove as additional means to prevent the tab from inadvertently sliding off of the peg **212**. Instead of using both the groove **244** and the rib **242**, one or the other of the groove **244** and the rib **242** may be used.

(25) As shown in FIGS. **6** and **10**, the horizontal peg **212** may be formed in a ledge **260** that extends around the container **210**, such as, for example, a ledge that is used to hold a cover (not shown) of the litter box **200**.

(26) FIG. **11** shows an embodiment of the tab **220** that can be used in the embodiments of FIG. **6-10**, which have the horizontally projecting peg **212**. The tab **220** includes a connection part **254** and a stiffener part **256**. The stiffener part **256** is similar to the stiffener part **122** described above. The upper portion **254** includes an opening for receiving the peg **212** and a grip portion **258** to facilitate removing the tab **220** from the peg **212**.

(27) Instead of being formed from a separate material and connected to the liner, the tab **120**, **220** may be incorporated into a liner as a die-cut portion of the material of the liner. In this alternative embodiment, a separate stiffener could be incorporated into the liner in the same manner as previously described, or the stiffener could also be made of the same material as the liner, i.e., by folding over a portion of the liner.

(28) The embodiments shown and described are four-sided litter boxes with front entry designs. However, the invention is not limited to such a design and can include a litter box of any shape. Also, the invention can also be applied to top-entry litter boxes instead of side entry boxes.

(29) Thus, while there has been shown and described and pointed out the fundamental novel features of the invention is applied to a preferred embodiment thereof, it will be understood that various omissions and substitutions and changes in the form and details of the devices illustrated, and in their operation, may be made by those skilled in the art without departing from the invention. For example, it is expressly intended that all combinations of those elements and/or method steps which perform substantially the same function in substantially the same way to achieve the same results are within the scope of the invention. Moreover, it should be recognized that structures and/or elements and/or method steps shown and/or described in connection with any disclosed form or embodiment of the invention may be incorporated in any other disclosed or described or suggested form or embodiment as a general matter of design choice. It is the intention, therefore, to be limited only as indicated by the scope of the claims appended hereto.

Claims

1. A litter box, comprising: a container having a wall; a plurality of pegs that project upward from a top of the wall or horizontally from an inner surface of the wall, the plurality of pegs being distributed along a circumference of the container; a liner that is insertable in the container, the liner having an upper edge; a plurality of tabs connected to the liner at or proximate the upper edge, each of the plurality of tabs having a connection part and a stiffener part, the connection part having an opening in which a respective peg of the plurality of pegs is receivable to hold the liner in place in the container, the stiffener part being connected to the liner, and the stiffener part extending laterally from opposing sides of the connection part and holding a portion of the upper edge of the liner against the wall of the container when the respective peg is received in the each of the plurality of tabs, wherein the plurality of pegs project upward from the top of the wall, and the each of the plurality of tabs includes a living hinge on the connection part, by which the opening of the each of the plurality of tabs can be pivoted to fit over the respective peg of the plurality of pegs.
2. The litter box according to claim 1, wherein the container has four wall portions corresponding

to four sides of the container and forming four corners, the four wall portions including a front wall portion, a rear wall portion, a first side wall portion, and a second side wall portion, and the plurality of pegs comprises four pegs arranged so that the plurality of tabs connected to the four pegs holds the liner in the four corners of the container.

3. The litter box according to claim 2, wherein a first pair of the four pegs is arranged on the first side wall portion and a second pair of the four pegs is arranged on the second side wall portion.

4. The litter box according to claim 3, wherein the four pegs are disposed so that one end of the stiffener part of the each of the plurality of tabs holds the liner in each of the four corners of the container.

5. The litter box according to claim 2, wherein the four corners are curved wall sections, and the four pegs are disposed at the curved wall sections of the four corners.

6. The litter box according to claim 1, wherein the connection part of the each of the plurality of tabs is divided into a first part and a second part by the living hinge, the first part being disposed between the living hinge and the stiffener part and the second part including the opening.

7. The litter box according to claim 1, wherein the each of the plurality of tabs includes a rib disposed on at least a portion of a border of the opening.

8. The litter box according to claim 1, wherein the each of the plurality of tabs includes a grip portion extending from a free end of the each tab.

9. The litter box according to claim 1, wherein the stiffener part of the each of the plurality of tabs is connected to the liner by stitching, friction fitting, gluing, welding, pinning, riveting, snap-fitting, and or crimping.

10. The litter box according to claim 1, wherein the stiffener part of the each of the plurality of tabs is sewn into a corresponding pocket of the liner.

11. The litter box according to claim 1, wherein the each of the plurality of tabs is die-cut from a portion of the same material as the liner.
